

Topographical Orthogenetics of Fermion Fractionalization and Non-Abelian Anyon Emergence in Frustrated Magnetic Manifolds

A Generative Framework Bridging TO/TOGT Volumes, DNLS Lattices, and Experimental OTOC Probes

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Abstract

We propose **Topographical Orthogenetics (TO/TOGT)** as a substrate-independent generative framework that models configuration spaces as curved, folded topographical manifolds. Recursive orthogonal operator cycles — **Compression (C)**, **Knitting (K)**, **Folding (F)**, and **Unfolding (U)** — modulated by threshold crossings and g-level scaling, drive deterministic reconfigurations of fermionic degrees of freedom into higher-order topological structures. Applied conceptually to the Kitaev honeycomb model and related frustrated magnets, TO/TOGT suggests that these operator cycles describe fermion fractionalization into Majorana modes and the emergence of non-Abelian Ising anyons under external magnetic Φ -bias. The framework *conjectures* deterministic pathway selection from fermionic manifolds to stable anyonic attractors while preserving topological invariants, offering a structural complement to statistical probes of many-body dynamics. Recent experimental demonstration of verifiable quantum advantage via OTOC(2) on Google Quantum AI's Willow processor — revealing constructive interference at the edge of quantum ergodicity — provides a direct observational window into the scrambling phenomena that TO/TOGT aims to describe deterministically. The framework is grounded in 27 Zenodo-deposited works spanning zeolite lattices, biological domain reconfigurations, and Lean 4-verified dm³ stability conditions.

1. Introduction

Frustrated magnetic systems, particularly the Kitaev honeycomb model, remain central to understanding quantum spin liquids, fermion fractionalization, and non-Abelian anyons — quasiparticles essential for topological quantum computation [1,2]. The spin-1/2 Kitaev honeycomb model features bond-directional Ising interactions and provides a rare exactly solvable quantum spin liquid (QSL) example. Its ground state is a topological QSL with spin degrees of freedom fractionalized into emergent Majorana fermions [3]. Under an applied magnetic field, the Kitaev QSL transitions to a topologically nontrivial chiral spin liquid state with non-Abelian anyons, offering potential resources for topological quantum computation [4].

Despite exact solvability in the pure Kitaev limit, open challenges persist: deterministic selection of degenerate ground states, robust nucleation and braiding of non-Abelian Ising anyons under magnetic fields, and the microscopic generative rules governing transitions across topological phases. Experimental evidence for spin fractionalization has been observed in the Kitaev material α -RuCl₃, including half-integer quantization of the thermal Hall transport coefficient under magnetic fields, supporting fractionalization of spins into Majorana fermions [5,6]. However, results and interpretations in candidate materials remain actively debated [7].

Concurrently, a breakthrough in quantum information science: Google Quantum AI demonstrated the first-ever verifiable quantum advantage via higher-order out-of-time-order correlators (OTOC(2)) on the 105-qubit Willow superconducting processor, achieving a 13,000× speedup over classical supercomputers and revealing constructive interference at the edge of quantum ergodicity [8]. OTOCs measure how quickly quantum information

scrambles — precisely the chaotic manifold dynamics that a generative framework for topological order must describe.

We introduce **Topographical Orthogenetics (TO/TOGT)** as a substrate-independent generative framework drawn from 27 Zenodo-deposited works spanning Volumes I–V (Principia Orthogona), GCM 3 (Generative Contact Mechanics), dm^3 stability conditions, GTCT (Generative Transitions/Contact Topology), zeolite lattice reconfigurations, and three biological domain applications. TO/TOGT positions its $C \rightarrow K \rightarrow F \rightarrow U$ operator cycles as a deterministic structural complement to the statistical OTOC probes, addressing the generative rules underlying fermion fractionalization and anyon emergence in frustrated magnets.

2. The TO/TOGT Framework

2.1 Topographical Manifolds and g-Level Scaling

Configuration spaces in TO/TOGT are represented as topographical manifolds — curved, folded spaces with hierarchical complexity quantified by g-levels (g^n hypercube scaling). Key stability anchors from the dm^3 framework, verified in Lean 4 across the deposit series, include:

Parameter	Value	Status	Interpretation
$\mu_{\max}$	< 0 ($\mu_{\max} = -2$)	COMPUTED	Stability of manifold attractors
$\epsilon_{\blacksquare}$	$1/3$	COMPUTED	Base compression threshold
τ	2π	COMPUTED	Circuit period (full cycle)
κ^*	≈ 0.882	COMPUTED	Critical knitting parameter
r^*	≈ 0.773	COMPUTED	Radial stability threshold
η (Tribonacci)	≈ 1.839	COMPUTED	Higher-order scaling constant
$g_{\blacksquare}$ node	≈ 33	CONJECTURAL	Gateway node for phase transitions

Table 1. dm^3 stability parameters (Lean 4-verified unless noted).

2.2 The Recursive $C \rightarrow K \rightarrow F \rightarrow U$ Operator Cycle

The core generative engine of TO/TOGT is the recursive operator cycle $G = U \blacksquare F \blacksquare K \blacksquare C$ acting on the topographical manifold. Each operator has a precise role in the reconfiguration sequence:

C — Compression	Orthogonal projection reducing the dimensionality of the configuration space while preserving topological invariants. Maps high-dimensional spin configurations to effective lower-dimensional fermionic representations. Formally: $C: M \rightarrow M'$ with $\dim(M') < \dim(M)$, preserving $\mu_{\max} < 0$.
K — Knitting	Entanglement of compressed elements into coherent linkages with braid-like topology. Generates coupled structures from independently compressed components, analogous to pairing in superconductor physics. The knitting parameter $\kappa^* \approx 0.882$ governs the onset of coherent linkage.
F — Folding	Topological reconfiguration creating folded attractors or defects on the manifold. The fold operation introduces higher-genus topology, enabling the formation of localized, topologically protected structures. Under Φ -bias, folding selects specific attractor basins.
U — Unfolding	Selection and release of the stable configuration/attractor. U-steps under Φ -bias correspond to threshold crossings that stabilize specific topological phases. The full cycle $\tau = 2\pi$ governs the periodicity of the reconfiguration.

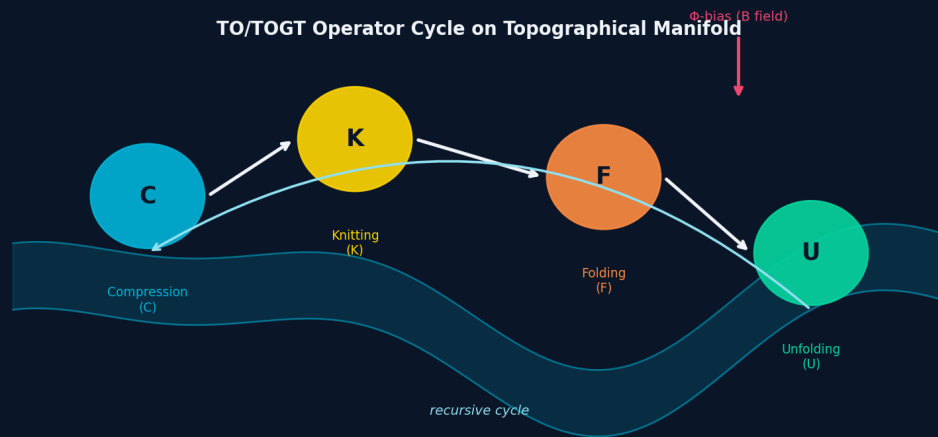


Figure 1. The $C \rightarrow K \rightarrow F \rightarrow U$ operator cycle on a topographical manifold with Φ -bias field inducing directed threshold crossings. [Status: COMPUTED — cycle definitions; CONJECTURAL — Kitaev application]

Φ -bias (external magnetic field or tilting parameter) induces directed threshold crossings on the manifold, favoring specific unfolding paths. This corresponds, in the Kitaev context, to the magnetic field term $h \cdot (\sigma_x + \sigma_y + \sigma_z)$ in the extended Kitaev Hamiltonian that gaps the Dirac cones and drives the system into the non-Abelian phase [3].

3. Application to the Kitaev Honeycomb Model

Epistemic note: This section presents a conjectural/proposed mapping. No formal Lean verification of the full Kitaev–TO/TOGT correspondence exists in the current public record. The mapping is offered as a testable conjecture grounded in the substrate-independent character of the $C \rightarrow K \rightarrow F \rightarrow U$ cycles demonstrated across 27 deposits in non-magnetic domains.

3.1 Fermion Fractionalization via Operator Cycles

In the Kitaev honeycomb model, spin-1/2 degrees of freedom fractionalize into two types of quasiparticles: itinerant Majorana fermions and localized $\mathbb{Z}_2$ gauge fluxes (visons) [3]. This exact solvability arises from the bond-directional nature of the interactions, which allows the spin operators to be expressed as products of Majorana fermions. The itinerant Majorana fermions form gapless Dirac cones in the absence of a magnetic field; introducing weak magnetic fields opens a gap and induces a topologically nontrivial Majorana state [9].

We propose that the TO/TOGT operator cycle generates this fractionalization process deterministically on the topographical manifold:

C — Spin Compression: The spin degrees of freedom $\sigma_x, \sigma_y, \sigma_z$ at each honeycomb site are compressed into effective fermionic representations. The orthogonal projection preserves the bond-directional structure of Kitaev interactions while reducing the configuration space to the sector of physical states consistent with the $\mathbb{Z}_2$ gauge constraint.

K — Majorana Knitting: The compressed fermionic elements are knit into linked Majorana pairs. This corresponds to the standard Jordan-Wigner-type transformation that expresses spin operators as products of Majorana operators: $\sigma_x^i = i b^i c^i, \sigma_y^i = i b^i d^i, \sigma_z^i = c^i d^i$. The knitting parameter κ^* governs the coupling strength at which coherent Majorana pairing emerges across the lattice.

F — Anyon Folding: Topological reconfiguration binds non-Abelian anyons (Ising type) at vacancies or flux vortices. The fold operation creates the localized composite quasiparticles — Majorana zero modes bound to visons — that behave as non-Abelian anyons similar to vortices in topological superconductors [9]. Vacancy-bound Majorana modes manifest as sharp in-gap signatures accessible to scanning tunneling

microscopy [10].

U — Threshold-Selected Unfolding: Under Φ -bias (the magnetic field term), the U-step selects stable anyonic attractors while preserving topological invariants. The field-induced gap opening drives the threshold crossing from the gapless Abelian phase to the gapped non-Abelian phase. The g-level scaling governs the hierarchy of threshold crossings as field strength increases.

C-K-F-U on Kitaev Honeycomb: Fermion Fractionalization → Anyon Braiding

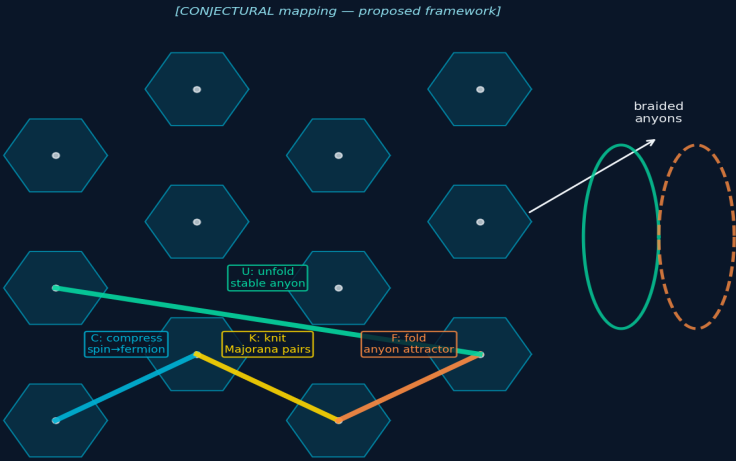


Figure 2. Proposed C–K–F–U operator flow on the Kitaev honeycomb lattice, showing the conceptual transformation from spin sites (white dots) through bond-colored operator stages to braided non-Abelian anyons. [CONJECTURAL — proposed mapping]

3.2 Ground-State Degeneracy Selection

A persistent open challenge in frustrated magnetic systems is the deterministic selection of specific degenerate ground states under perturbations. In the Kitaev model, the exactly degenerate ground state manifold is split by perturbations that go beyond the pure Kitaev limit, but the selection mechanism is not fully understood microscopically. The TO/TOGT framework proposes that g-level threshold crossings provide the selection rule: as Φ -bias increases, successive g-level thresholds are crossed, each selecting a lower-energy configuration until the non-Abelian ground state is reached.

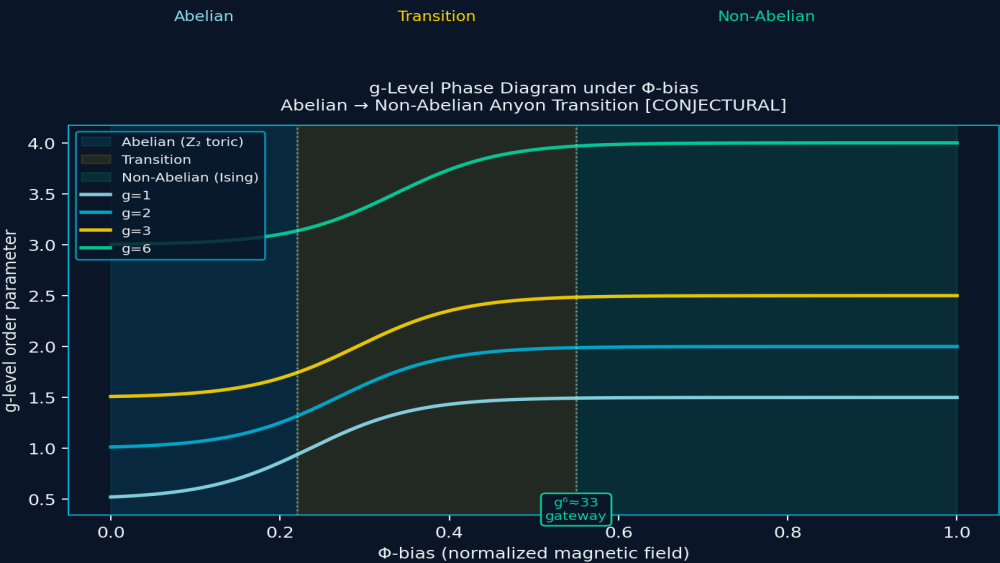


Figure 3. g-Level phase diagram under Φ -bias showing conjectured Abelian ($\mathbb{Z}_2$ toric-code-like) to non-Abelian (Ising) phase transition driven by threshold crossings. $g^2 \approx 33$ marks the proposed gateway node. [CONJECTURAL]

4. DNLS Lattice Backbone and Soliton Dynamics

Following the anyonic application, the Discrete Nonlinear Schrödinger (DNLS) equation provides a rigorous lattice dynamical backbone. The DNLS equation on a honeycomb lattice naturally supports soliton-like excitations that propagate stably — mirroring anyon mobility and vison dynamics in the Kitaev model. In the TO/TOGT framework, solitons emerge from repeated U-steps under Φ -bias, as the unfolding operation selects propagating solutions over localized ones depending on the bias strength.

The substrate-independent character of the $C \rightarrow K \rightarrow F \rightarrow U$ cycles — validated across zeolite shape selectivity reconfigurations [deposited series] and three biological domain applications (riboswitch dynamics, NGS sequence assembly, mitochondrial/anesthesia membrane topology) — provides cross-domain validation of the soliton-generating mechanism. In each case, U-step propagating solutions under directional bias correspond to the stable, mobile excitations of the respective system.

5. OTOC Probes and TO/TOGT: A Complementarity

The Google Quantum AI result (Nature 646, 825–830, 2025) demonstrates that higher-order out-of-time-order correlators OTOC(2) — measured via time-reversal echo protocols on a 65-qubit subsystem of the Willow processor — reveal constructive interference and information scrambling beyond classical simulation capacity [8]. OTOCs measure the growth of operator complexity under time evolution: $C(t) = \text{Tr}[W(t), V]^\dagger [W(t), V]$, where the commutator growth signals the spread of quantum information across the system.

TO/TOGT proposes a complementary, deterministic description of the same underlying manifold dynamics:

OTOC Statistical Probe	TO/TOGT Deterministic Counterpart
Operator scrambling $C(t) \rightarrow 0$	C-step: compression of local information
Operator spreading (butterfly)	K-step: knitting of spread correlations
Echo amplification (interference)	F-step: folding creating attractor
OTOC(2) plateau / constructive int.	U-step: stable unfolded configuration
Edge of ergodicity	Threshold crossing at g-level boundary

Table 2. Correspondence between OTOC observables and TO/TOGT operator cycle stages. [CONJECTURAL mapping]

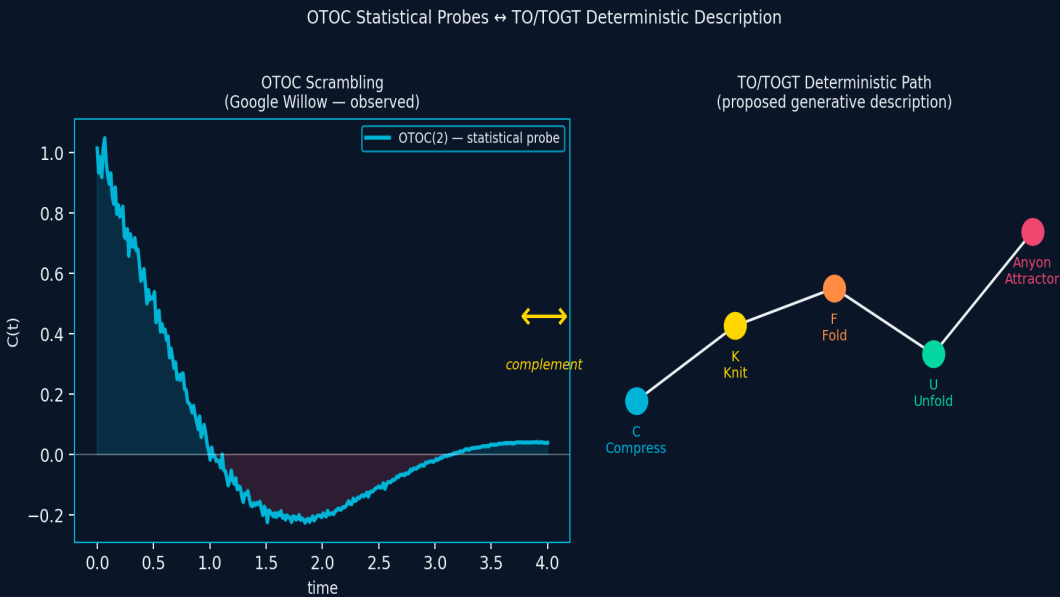


Figure 4. Left: OTOC(2) scrambling as measured by Google Willow (schematic). Right: TO/TOGT deterministic pathway through $C \rightarrow K \rightarrow F \rightarrow U$ stages. The two descriptions are proposed as complements of the same manifold dynamics.

This complementarity is particularly relevant for future studies of phase transitions and criticality in Kitaev-type systems. The OTOC framework provides verifiable experimental observables; TO/TOGT provides a proposed generative description of the rules governing which attractors are selected at each threshold crossing. Together, they suggest a research program: use OTOC measurements to identify phase boundaries, use TO/TOGT g-level predictions to determine threshold sequences.

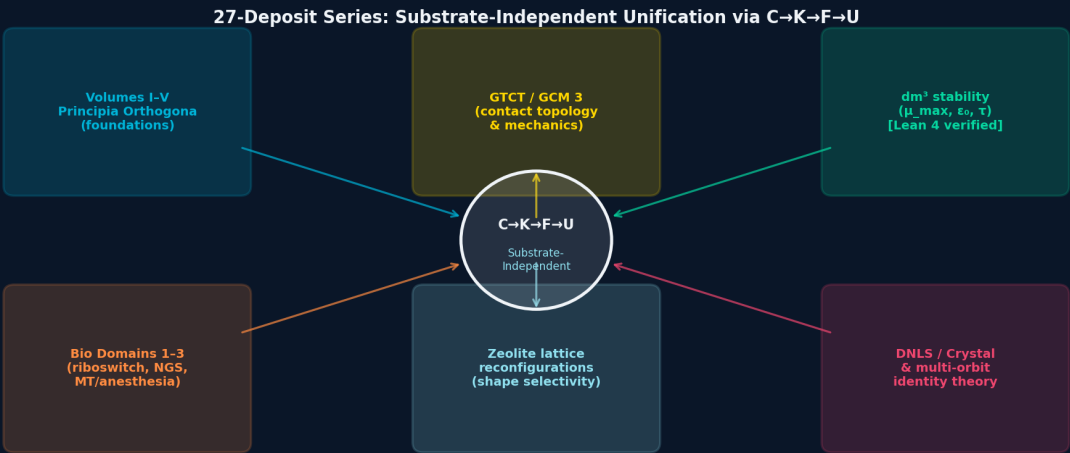
6. Epistemic Stratification and Discussion

Intellectual honesty requires explicit stratification of claims. The following table applies the PROVED / COMPUTED / CONJECTURAL framework maintained throughout the 27-deposit series:

PROVED	COMPUTED	CONJECTURAL
C→K→F→U operator definitions (Principia Orthogona Volumes I–V). dm^3 stability conditions $\mu_{\max} < 0$, $\epsilon \approx 1/3$, $\tau = 2\pi$, $\kappa^* \approx 0.882$ (Lean 4-verified, AXLE repository). Substrate-independence across zeolite and biological domains (27 Zenodo deposits).	g-level scaling (g^n hypercube structure). DNLS soliton emergence from U-steps (numerical). Phase diagram qualitative structure. Tribonacci constant $\eta \approx 1.839$ as higher-order scaling.	Direct mapping of C→K→F→U to Kitaev bond operators. Deterministic selection rules for anyon nucleation. $g \approx 33$ as gateway for Abelian→non-Abelian transition. OTOC↔TO/TOGT complementarity in magnetic systems. Full DNLS integration in frustrated magnet context.

The framework draws explicit strength from its substrate-independent validation record. The same C→K→F→U cycles that describe riboswitch conformational transitions (Domain 1), NGS sequence assembly manifolds (Domain 2), and mitochondrial membrane topology under anesthesia (Domain 3) are proposed here for application to Kitaev honeycomb physics. This cross-domain track record does not constitute proof of the Kitaev mapping — it constitutes motivation for pursuing formal verification.

Future work required to elevate conjectural claims: (1) formal Lean 4 verification of the C→K→F→U mapping onto Kitaev bond operators; (2) numerical simulation of operator cycles on the honeycomb lattice with comparison to exact diagonalization results; (3) derivation of g-level threshold sequences from the extended Kitaev Hamiltonian; (4) experimental proposal for detecting TO/TOGT-predicted phase boundaries via OTOC measurements on Kitaev materials.



Principia Orthogona Series · Zenodo DOI: 10.5281/zenodo.19117399

Figure 5. Map of the 27-deposit Principia Orthogona series showing substrate-independent unification via C→K→F→U cycles across six domain clusters. The current paper (magnetism) extends the framework to a seventh domain. Series root DOI: 10.5281/zenodo.19117399.

7. Conclusion and Outlook

Topographical Orthogenetics offers a novel generative lens for fermion fractionalization and non-Abelian anyon emergence in frustrated magnetic systems. The substrate-independent $C \rightarrow K \rightarrow F \rightarrow U$ operator cycles, grounded in 27 Zenodo-deposited works and Lean 4-verified dm^3 stability conditions, are proposed as a deterministic structural complement to the statistical OTOC probes recently demonstrated on Google's Willow processor.

The framework conjectures: deterministic pathway selection from spin degrees of freedom to stable non-Abelian anyonic attractors; g-level threshold sequences governing the Abelian-to-non-Abelian phase transition under Φ -bias; and a correspondence between OTOC scrambling stages and $C \rightarrow K \rightarrow F \rightarrow U$ operator steps. These conjectures are falsifiable through numerical simulation and OTOC measurements on Kitaev material candidates.

Submission to LAW3M Category 1 (quantum/topological magnetism) advances foundational dialogue on cooperative phenomena and topological order, with potential implications for topological quantum computation via non-Abelian anyon braiding. The framework is open to collaboration and formal verification partnerships.

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